STA2023 – Statistical Methods I
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Chapter 5. Random Variables

Table of Contents

5.1 Discrete Random Variables 2

Discrete vs. Continuous Random Variables 2

Discrete Probability Distributions 3

Mean of a Discrete Random Variable 5

Finding the Standard Deviation of a Discrete Random Variable 8

5.2 The Binomial Probability Distribution 9

Computing Binomial Probabilities 10

Using Excel for Binomial Distribution: ..... **Error! Bookmark not defined.**

Mean and Standard Deviation of a Binomial Random Variable 13

Learning Objectives: By the end of this chapter, students should be able to:

- Identify discrete and continuous random variables.
- Create and validate discrete probability distributions
- Find the mean and standard deviation of discrete probability distributions.
- Identify binomial experiments.
- Use the binomial distribution for probability calculations.
- Find the mean and standard deviation of a binomial random variable.

5.1 Discrete Random Variables

Discrete vs. Continuous Random Variables

In chapter 4, we talked about various probability rules for different events. In chapter 5, we are going to consider probability events again, but now we are only considering probability experiments with numeric outcomes.

Definition: A **random variable**

is a numerical measure of the outcome of a probability experiment; its value is determined by chance. Written with a capital letter (X); a specific value is written with a small letter (x).

A **discrete random variable**

has a finite or countable number of possible values — you can LIST them (0, 1, 2, 3, …); they are separate points on the number line. Usually comes from COUNTING.



A **continuous random variable**

has infinitely many possible values — any value in an interval (a whole segment of the number line). Comes from MEASURING (time, weight, length, …).



Example 1: A group of students plan to purchase a 16-ounce bag of M&M's and record data on the contents. For each of the following variables, state whether it is discrete or continuous. State possible values for the random variable.

1. Mass of each of the M&M's

   Continuous — possible values: m > 0 (any value in an interval of masses)

2. Number of pieces of candy in the bag

   Discrete — possible values: x = 0, 1, 2, 3, … (a count)

3. Diameter of each piece of candy

   Continuous — possible values: d > 0 (any value in an interval of lengths)

4. Number of distinct colors of M&M's in the bag

   Discrete — possible values: x = 1, 2, 3, 4, 5, 6

Discrete Probability Distributions

Definition: The **probability distribution** of a discrete random variable X

provides the possible values of X and their corresponding probabilities. It can be shown as a TABLE, a GRAPH (probability histogram), or a FORMULA.

Rules for Discrete Probability Distribution:

1. Every probability is between 0 and 1: $0 \le P(x) \le 1$
2. The probabilities add up to 1: $\Sigma P(x) = 1$
(To "verify" a distribution, check both rules.)

Example 2: At insurance companies, actuaries use various data sources, tables, and models to determine various probabilities of needing to pay out certain policies. Suppose that the probability that a healthy 24-year-old will die in a given year is 0.0014. The insurance company will pay out $10,000 to the beneficiaries, but the policy costs $100 per year.

Let X be the amount of profit earned by the insurance company in a year for a 24-year-old policy holder. Create and verify the discrete probability distribution for X.

Lives (probability 1 − 0.0014 = 0.9986): profit = +$100. Dies (probability 0.0014): profit = $100 − $10,000 = −$9,900.
x = $100 → P(x) = 0.9986 x = −$9,900 → P(x) = 0.0014
Verify: both probabilities are between 0 and 1 ✓ and 0.9986 + 0.0014 = 1 ✓ → valid distribution.

Example 3: At a monthly 'casino night,' there is a game called Chuck-a-Luck: Three dice are rolled in a wire cage. You place a bet on any number from 1 to 6. If any one of the three dice comes up with your number, you win the amount of your bet. (You also get your original stake back.)

If more than one die comes up with your number, you win the amount of your bet for each match. For example, if you had a $1 bet on number 5, and each of the dice came up with 5, you would win $3. There are four possible outcomes. (The selected number can match 0, 1, 2, or 3 of the dice.)

The random variable *X* represents the profit from a $1 bet in Chuck-A-Luck. The table below summarizes the probabilities of earning a profit of *x* dollars from a $1 bet. Verify that this is a discrete probability distribution.

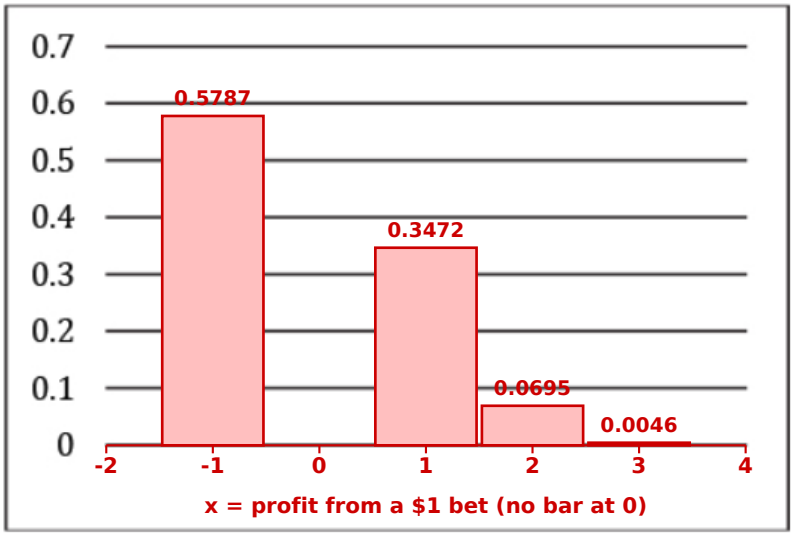
Number of Dice Matching the Chosen Number	Profit	Probability
0	-$1	0.5787
1	$1	0.3472
2	$2	0.0695
3	$3	0.0046

Each probability is between 0 and 1 ✓. Sum: 0.5787 + 0.3472 + 0.0695 + 0.0046 = 1.0000 ✓
→ YES, it is a discrete probability distribution.

- Probability distributions can also be shown via graphs. For example, with the example above:

Profit	Probability
-$1	0.5787
$1	0.3472
$2	0.0695
$3	0.0046

Probability histogram: one bar centered on each value of x (width 1), bar height = P(x) — drawn in red below



Mean of a Discrete Random Variable

Suppose we have a random variable X. What would a typical value for X be? For example, if we play the dice game, do we expect to make a profit or lose money, on average?

Definition: The **mean of a discrete random variable** is given by the formula:

$\mu_X = \Sigma [x \cdot P(x)]$ — multiply each value by its probability, then add them all up.
Interpretation: the average value of X if the experiment were repeated many, many times (the long-run average).

Example 4: The random variable X represents the profit from a \$1 bet in Chuck-A-Luck. The table below summarizes the probabilities of earning a profit of x dollars from a \$1 bet. Compute and interpret the mean of the random variable X.

Profit	Probability
-\$1	0.5787
\$1	0.3472
\$2	0.0695
\$3	0.0046

$\mu_X = \Sigma\, x \cdot P(x) = (-1)(0.5787) + (1)(0.3472) + (2)(0.0695) + (3)(0.0046)$
$= -0.5787 + 0.3472 + 0.1390 + 0.0138$
$= -0.0787 \approx -\$0.08$
Interpretation: if you played Chuck-a-Luck many, many times betting \$1 each time, you would LOSE about 8 cents per game on average (the casino wins in the long run).

Note: When interpreting means, it's important to note that these averages are in the extreme long run (if you repeated this an infinite number of times!).

Definition: The mean of a random variable represents what we would expect to happen in the long run, so it is also called the expected value, or E(X). The interpretation of expected value is the same as the interpretation of the mean of a discrete random variable.

Example 5: Refer back to the insurance table. The amount of distribution of X, the amount of profit for the insurance company in a year, is shown below.

Profit	Probability
$100	.9986
-$9900	.0014

Find the expected value of the profit for the insurance company. Should the insurance company continue selling this policy?

$E(X) = \Sigma\, x \cdot P(x) = (100)(0.9986) + (-9900)(0.0014) = 99.86 - 13.86 = \$86.00$
The company expects to earn an average profit of $86 per policy per year in the long run → YES, keep selling the policy (positive expected value).

Finding the Standard Deviation of a Discrete Random Variable

- The expected value finds the "average" or "typical" result of a discrete random variable, but how much will it vary from experiment to experiment?
- We can use standard deviation to explain that:

Definition: The standard deviation of a discrete random variable X is given by:

$\sigma_X = \sqrt{\Sigma\,[\,(x-\mu_X)^2 \cdot P(x)\,]}$ (shortcut: $\sigma_X = \sqrt{\Sigma\,[\,x^2 \cdot P(x)\,] - \mu_X^2}$)
Steps: (1) find μ_X; (2) for each x: $(x-\mu_X)^2$, then multiply by P(x); (3) add them up = variance σ^2; (4) square root.

Example 6: Refer to the Chuck-A-Luck example. Computer the standard deviation of the random variable X.

Profit	Probability
-$1	0.0587
$1	0.3472
$2	0.0695
$3	0.0046

(Typo in the table: P(−$1) should be 0.5787, as in Examples 3–4.) $\mu_X = -0.0787$ from Example 4.
$(-1-(-0.0787))^2 \cdot 0.5787 = (-0.9213)^2 \cdot 0.5787 = 0.8488 \cdot 0.5787 = 0.4912$
$(1+0.0787)^2 \cdot 0.3472 = (1.0787)^2 \cdot 0.3472 = 1.1636 \cdot 0.3472 = 0.4040$
$(2+0.0787)^2 \cdot 0.0695 = (2.0787)^2 \cdot 0.0695 = 4.3210 \cdot 0.0695 = 0.3003$
$(3+0.0787)^2 \cdot 0.0046 = (3.0787)^2 \cdot 0.0046 = 9.4784 \cdot 0.0046 = 0.0436$
$\sigma^2 = 0.4912 + 0.4040 + 0.3003 + 0.0436 = 1.2391$ $\sigma_X = \sqrt{1.2391} \approx 1.1131 \approx \$1.11$
(Shortcut check: $\Sigma\, x^2P(x) = 0.5787 + 0.3472 + 0.2780 + 0.0414 = 1.2453$; $1.2453 - (-0.0787)^2 = 1.2391$ ✓)

5.2 The Binomial Probability Distribution

- Certain discrete random variables all follow very similar patterns. One of these is when we want to count the number of "successes" in a number of trials, where each trial only has two results: a "success" or a "failure".
    - For example, if you throw 10 free throws and the probability you make a free throw on any given throw is 90%, what is the probability you make 8 of the 10 free throws?
- This calculation can be tricky since we would have to list out all the ways you could get 8 successes out of 10. However, since it follows a certain pattern, we can use that pattern to make probability calculations easier.
- **Definition:** An experiment is said to be a **binomial experiment** if

all FOUR conditions below hold. X = number of successes in the n trials (x = 0, 1, …, n).

1. **The experiment is performed a FIXED number of times, n (each repetition is a trial).**
2. **The trials are INDEPENDENT — the outcome of one trial does not affect the others.**
3. **Each trial has only TWO possible outcomes: "success" or "failure".**
4. **The probability of success, p, is the SAME for every trial.**

Example 7: Which of the following is a binomial experiment? For any binomial experiment, identify the values of n, p, and x (the potential number of successes).

1. A player rolls a pair of fair die 10 times. The number X of 7's rolled is recorded.

YES, binomial: fixed n = 10 rolls; rolls are independent; each roll is either "a 7" (success) or "not a 7" (failure); p = P(sum of 7) = 6/36 = 1/6 ≈ 0.1667 is the same on every roll.
n = 10, p = 1/6, x = 0, 1, 2, …, 10

2. The proportion of flights that arrived on-time according to the U.S. Department of Transportation in July 2016 was 0.7515. In order to assess reasons for delays, an official with the FAA randomly selects flights from July 2016 until she finds 10 that were not on time. The number of flights X that need to be selected is recorded.

NOT binomial: the number of trials is not fixed — she keeps selecting flights UNTIL she finds 10 late ones (the number of successes is fixed instead of n).

3. In a class of 30 students, 55% are female. The instructor randomly selects 4 students. The number X of females selected is recorded.

NOT binomial: 4 students are chosen WITHOUT replacement from only 30, so the trials are not independent — p changes after each pick (4/30 = 13% of the population, more than 5%, so independence can't be assumed).

Computing Binomial Probabilities

According to the Bureau of Transportation Statistics, in April, 2023 the proportion of on-time flights at Las Vegas's McCarran International Airport was 0.70. Suppose that 4 flights are randomly selected from April, 2023 and the number of on time flights X is recorded.

n = 4 trials, success = on time, p = 0.70, failure = 1 − p = 0.30. X = number on time = 0, 1, 2, 3, 4.
Any ONE sequence with x on-time flights has probability (0.70)^x (0.30)^(4 − x); there are 4Cx such sequences, so:
P(X = 0) = 4C0 (0.70)^0 (0.30)^4 = 1 · 0.0081 = 0.0081
P(X = 1) = 4C1 (0.70)^1 (0.30)^3 = 4 · 0.70 · 0.027 = 0.0756
P(X = 2) = 4C2 (0.70)^2 (0.30)^2 = 6 · 0.49 · 0.09 = 0.2646
P(X = 3) = 4C3 (0.70)^3 (0.30)^1 = 4 · 0.343 · 0.30 = 0.4116
P(X = 4) = 4C4 (0.70)^4 (0.30)^0 = 1 · 0.2401 = 0.2401
Check: 0.0081 + 0.0756 + 0.2646 + 0.4116 + 0.2401 = 1.0000 ✓ e.g. P(exactly 3 on time) = 0.4116

Definition: Binomial Probability Distribution Function:

P(x) = nCx · p^x · (1 − p)^(n − x), x = 0, 1, 2, …, n
n = number of trials, p = probability of success on one trial, 1 − p = probability of failure,
x = number of successes, nCx = n! / [x! (n − x)!] = number of ways to place the x successes among the n trials.
Excel: exactly x → =BINOM.DIST(x, n, p, FALSE) P(X ≤ x) → =BINOM.DIST(x, n, p, TRUE)
Steps: (1) write n, p, 1 − p; (2) translate the words into a symbol (table below); (3) add up P(x) for every x that fits, or use the complement.

Phrase	Math Symbol
“at least” or “no less than” or “greater than or equal to”	≥
“more than” or “greater than”	>
“fewer than” or “less than”	<
“no more than” or “at most” or “less than or equal to”	≤
“exactly” or “equals” or “is”	=

P(X ≤ k) = P(0) + P(1) + … + P(k)
P(X < k) = P(X ≤ k − 1)
P(X > k) = 1 − P(X ≤ k)
P(X ≥ k) = 1 − P(X ≤ k − 1) “at least 1” = 1 − P(0)

Example 8: According to CTIA, 72% of all adult Americans would rather give up chocolate than their cell phone. In a random sample of 10 adult Americans, what is the probability that

(a) Exactly 8 would rather give up chocolate?

(b) Fewer than 3 would rather give up chocolate?

(c) At least 3 would rather give up chocolate?

n = 10, p = 0.72, 1 − p = 0.28, X = number who would rather give up chocolate.

(a) "Exactly 8" → P(X = 8) = 10C8 (0.72)^8 (0.28)^2 = 45 · 0.07222 · 0.0784 ≈ 0.2548

(b) "Fewer than 3" → P(X < 3) = P(0) + P(1) + P(2)
P(0) = 10C0 (0.72)^0 (0.28)^10 = 1 · 1 · 0.000002962 = 0.0000030
P(1) = 10C1 (0.72)^1 (0.28)^9 = 10 · 0.72 · 0.00001058 = 0.0000762
P(2) = 10C2 (0.72)^2 (0.28)^8 = 45 · 0.5184 · 0.00003778 = 0.0008813
P(X < 3) = 0.0000030 + 0.0000762 + 0.0008813 = 0.0009605 ≈ 0.0010

(c) "At least 3" → P(X ≥ 3) = 1 − P(X < 3) = 1 − 0.0009605 = 0.9990395 ≈ 0.9990
(complement rule — much faster than adding P(3) through P(10))

Example 9: Suppose that you are pretty good at basketball and can make a free throw 95% of the time. Suppose you throw 3 free throws.

What is the probability that:

(a) You make all 3 free throws?

(b) You make less than 3 free throws?

n = 3, p = 0.95, 1 − p = 0.05, X = number of free throws made.
(a) P(X = 3) = 3C3 (0.95)^3 (0.05)^0 = 1 · 0.857375 · 1 ≈ 0.8574
(b) "Less than 3" = not all 3 → P(X < 3) = 1 − P(X = 3) = 1 − 0.8574 = 0.1426
(check: P(0)+P(1)+P(2) = 0.000125 + 0.007125 + 0.135375 = 0.1426 ✓)

Mean and Standard Deviation of a Binomial Random Variable

- If a random variable follows the binomial distribution, then the mean and standard deviation both have fairly nice formulas.

Definition: A binomial experiment with n independent trials and probability of success p has a mean and standard deviation given by the formulas

μX = n · p **σX = √(n · p · (1 − p))**

Example 12: According to flightstats.com, American Airlines flights from Dallas to Chicago are on time 80% of the time. Suppose 15 flights are randomly selected, and the number of on-time flights is recorded. Find the mean and standard deviation of the number of flights of the 15 that are on time.

n = 15, p = 0.80: μX = 15 (0.80) = 12 on-time flights (on average)
σX = √(15 · 0.80 · 0.20) = √2.4 ≈ 1.549 ≈ 1.55 flights